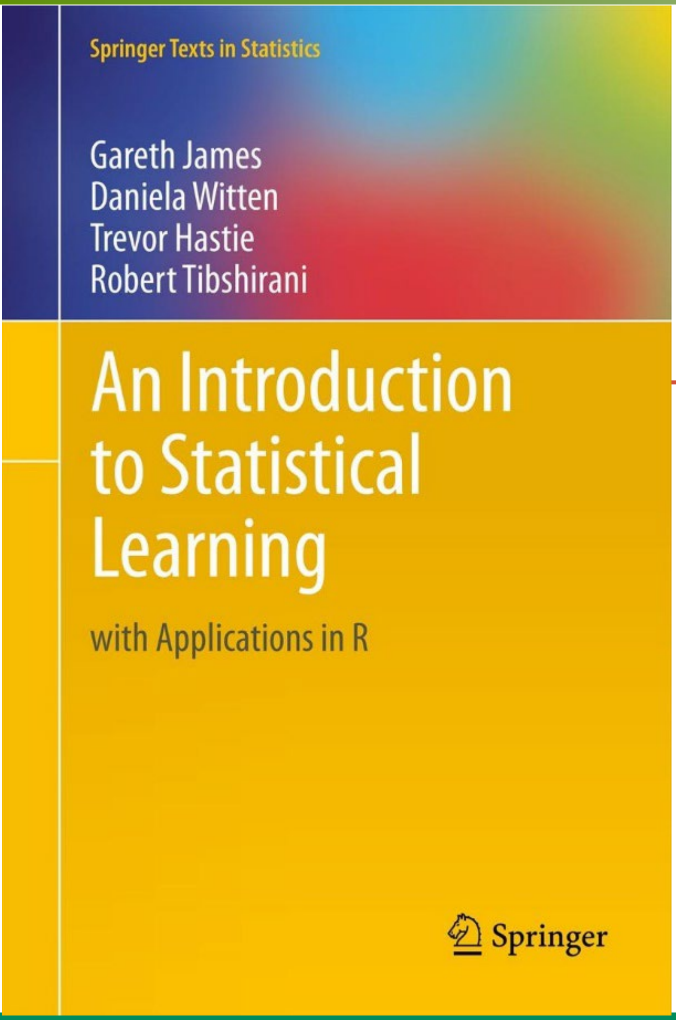

ANALYTICS

Week 14-15

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Examples Inspired By SOA, CAS, Coaching Actuaries ©

Key Formulas This Week

- Poisson: $P(y=j) = e^{-\mu}\frac{\mu^j}{j!}$, $E[y] = \mu = Var(y)$
- Log-link Offset: $\ln \mu_i = \ln E_i + \boldsymbol{x}_i'\boldsymbol{\beta}$
- Poisson $e_{i,Pearson} = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$
- Poisson $l(\boldsymbol{b}) = \sum_{i=1}^{n}[y_i \ln \mu_i - \mu_i - \ln(y_i!)]$
- Score Equations: $\sum_{i=1}^{n} e_i = 0$ and $\sum_{i=1}^{n} x_i e_i = 0$
- Poisson $D = 2\sum_{i=1}^{n}\left[y_i \ln\left(\frac{y_i}{\hat{\mu}_i}\right) - (y_i - \hat{\mu}_i)\right]$
- Poisson Overdispersion: $\hat{\varphi} = \frac{1}{n-(p+1)}\sum_{i=1}^{n}\frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}$

Key Formulas This Week 2

- NegBin: $P(y=j) = \binom{j+r-1}{r-1} p^r (1-p)^j$, $E[y] = \frac{r(1-p)}{p}$, $Var(y) = \frac{r(1-p)}{p^2}$
- NegBin Log-Link: $\mu_i = \frac{r(1-p_i)}{p_i} = e^{\boldsymbol{x}_i'\boldsymbol{\beta}}$
- Zero-Inflated: $P(y_i = j) = \begin{cases} \pi_i + (1-\pi_i)g_i(0), j = 0 \\ (1-\pi_i)g_i(j), j = 1,2,\ldots \end{cases}$, $E[y_i] = (1-\pi_i)\mu_i$, $Var(y_i) = (1-\pi_i)\mu_i + \pi_i(1-\pi_i)\mu_i^2$
- Hurdle: $P(y_i = j) = \begin{cases} \pi_i, j = 0 \\ k_i g_i(j), j = 1,2,\ldots \end{cases}$, $k_i = \frac{1-\pi_i}{1-g_i(0)}$, $E[y_i] = k_i\mu_i$, $Var(y_i) = k_i\mu_i + k_i(1-k_i)\mu_i^2$

Count Dependent

- A discrete count dependent variable y takes on non-negative integer values 0, 1, 2, etc.

- In actuarial applications, the most common use is **frequency** modeling for the ___

Poisson Distribution

- $P(y = j) = e^{-\mu} \frac{\mu^j}{j!}$ for $j = 0,1,2, \ldots$
- $=$
- in the linear exponential family format $f(y; \theta, \varphi) = e^{\left[\frac{y\theta - b(\theta)}{\varphi} + S(y,\varphi)\right]}$
- $\theta =$
- $b(\theta) =$
- $\varphi = 1$
- $S(y, \varphi) =$
- $E[y] =$

Poisson Coefficients

- Using a Poisson GLM with log link, the estimated mean changes by a factor of e^{b_j} per _____
- ie. $\ln \hat{y} = 0.5 - 0.1x_1 + 0.2x_2$
- Thus, the sign of b_j indicates correlation with y.

Poisson Exposures

- In the Poisson model, the # of **exposures**, which represents the scalable ___
- $E[y_i] =$
  - μ: Mean count per exposure
  - E_i: # of exposures for risk i
- Observed counts cannot be directly compared because their exposures may differ.

Poisson Offset

- The log link function is typically used.
- The **offset** term accounts for varying exposures and impacts the ___
- $\ln \mu_i =$
- β_j is the proportional change in the mean per unit change in x_{ij} if the log-link is used.

Poisson Model Properties

- Under Poisson regression, the maximum likelihood estimate for $\hat{\mu}$ is ____

- Score Equations:

Poisson Model Properties 2

- $l(\boldsymbol{b}) =$
- $D =$
- $d_{i,Deviance} = \text{sign}\,(y_i - \hat{\mu}_i)\sqrt{D_i}$

Statistical Tests

- For testing $H_0: \beta_j = 0$ against $H_a: \beta_j \neq 0$ and in determining confidence intervals, use the z-Table for GLMs, not the t-Table as for multiple linear regression.
- $z =$
- For a 2-sided z-test, the p-value is $2 * P(Z > |z|)$.

Poisson GLM Diagnostics

- These follow the same concepts as GLMs.
- $e_{i,Pearson} = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$
- $R^2_{maxscaled} = \frac{1 - e^{\frac{2(l_0 - l)}{n}}}{1 - e^{\frac{2 l_0}{n}}}$
- $R^2_{pseudo} = \frac{l_{\hat{\theta}} - l_0}{l_{SAT} - l_0}$
- $LRT = 2(l_a - l_0) = D_0 - D_a \sim \chi^2_q$

1. Theory Illustration

- A Poisson GLM with log link is built on the following dataset:

y_i	**5**	**2**	**1**
x_i	0	1	1

- The maximum likelihood estimate of the intercept is 1.6094.
- A. Calculate the coefficient of the predictor using MLE.
- B. Find the model maximized log-likelihood.

1. Theory Illustration 2

- Score Equations: $\sum_{i=1}^{n} e_i = 0$ and $\sum_{i=1}^{n} x_i e_i = 0$
- Log-likelihood function: $l(\boldsymbol{b}) = \sum_{i=1}^{n} [y_i \ln \mu_i - \mu_i - \ln(y_i!)]$
- $\ln \mu_i = b_0 + b_1 x_i$

Example 1

- A Poisson GLM with a log link models the $s (integers) of individual spending on video games.
- Calculate the predicted spending amount and its standard error for a group of ten 30-year old females. Interpret the Female coefficient.
- If the actual total spent by the group was $250, calculate the Pearson residual. Is it an outlier?

Parameter	b	$se(b)$
Intercept	2	0.2
Gender: Male	0	0
Gender: Female	-0.5	0.15
Age	0.05	0.02

Example 1 Sol

- $\ln \mu_i = \ln E_i + \boldsymbol{x}_i'\boldsymbol{\beta}$
- $r_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$

Overdispersion

- Data **overdispersion**, where ____

- Equi-dispersion: Variance = Mean

- Under-dispersion: Variance < Mean

Overdispersion 2

- To mitigate over/under-dispersion and still use Poisson models, let
- $Var(y_i) =$
  - φ: Dispersion parameter
- $\boldsymbol{b}$ is still estimated the same way but becomes a ____

Overdispersion 3

- The estimated dispersion parameter from the data is:
- $\hat{\varphi} =$
- While the estimated mean is calculated the same way, it can no longer be used to estimate probabilities, as the ___

Concept Check #1

- Which statements are true?
- I. Accounting for varying exposures will change the Poisson model coefficient estimates.
- II. Accounting for varying exposures helps handle over-dispersion.
- III. Poisson regression is suited for modeling aggregate claims.
- IV. Regression coefficients for a Poisson model can estimate variance of the response.

2. Test Illustration

- A **quasi-Poisson** GLM w/log-link models video game spend.
- $\sum_{i=1}^{100} \frac{(y_i - \hat{y}_i)^2}{\hat{y}_i} = 145.5$
- A. What is the % change in the estimated mean from 2 additional exposures of 30-year old females to an original of 10?
- B. Calculate the p-value for testing $H_a: b_{Age} \neq 0$ after adjusting for overdispersion.

Parameter	$\boldsymbol{b}$	$\boldsymbol{se(b)}$
Intercept	2	0.2
Gender: Male	0	0
Gender: Female	-0.5	0.15
Age	0.05	0.02

2. Test Illustration

- $Var(y_i) = \varphi\hat{\mu}_i$
- $\hat{\varphi} = \frac{1}{n-(p+1)}\sum_{i=1}^{n}\frac{(y_i-\hat{\mu}_i)^2}{\hat{\mu}_i} = \frac{\sum_{i=1}^{n} r_i^2}{n-(p+1)}$
- $\ln\mu_i = \ln E_i + \boldsymbol{x}_i'\boldsymbol{\beta}$
- $z = \frac{b_j - b_j(H_0)}{\sqrt{\hat{\varphi}} * se(b_j)}$

Negative Binomial Model

- Another popular count model uses the negative binomial distribution, with density function:
- $P(y = j) = \binom{j+r-1}{r-1} p^r (1-p)^j$
  - r, p are the negative binomial parameters
- $E[y] =$
- $Var(y) =$

Negative Binomial Model 2

- A nice interpretation of the negative binomial is that it is a mixture of Poisson variables.
- Poisson-Gamma Mixture: If $Y|\lambda$ is Poisson with mean λ, and λs follows a gamma distribution, then Y is a ____
- A log-link is typically used.
- $\mu_i = \frac{r(1-p_i)}{p_i} = e^{\boldsymbol{x}_i'\boldsymbol{\beta}}$
- Typically, r is chosen such that $\sigma =$

Models Summary

- Use Poisson/Negative Binomial regression for count response (ie. Frequency)

- Use a GLM with Tweedie distribution for a response with a discrete mass at 0 and continuous above.

- Use MLR only with a normally distributed response.

Latent Class Model

- **Latent** classification information represents innate but unknown information that exists.
- A latent class model uses a discrete mixture involving an unknown discrete random variable.
- Assuming 2 classes, with π_L being probability of being in the low-risk and π_H for high risk,

- $P(y_i = j) =$

Heterogeneity Model

- A heterogeneity model is a continuous mixture model.
- A random parameter α_i represents the **heterogeneity component**, attempting to capture unobserved features.
- $E[y_i|\alpha_i] = e^{\alpha_i + \boldsymbol{x}_i' \boldsymbol{\beta}}$ when log-link is used
- $E[y_i] = \mu_i$ with $E[e^{\alpha_i}] = 1$ assumed
- A negative binomial model is an example of a heterogeneity model.

Zero-Inflated Model

- Due to the high presence of 0s in insurance count data that few distributions can naturally handle, let Y be the mixture of a point mass at 0 (with probability π_i) and another distribution $g_i(j)$.

- $P(y_i = j) =$

- This is a special case of a latent class model.

Zero-Inflated Model 2

- π_i is often modeled using a binary model.
- Assuming g_i is Poisson with mean μ_i,
- $E[y_i] =$
- $Var(y_i) =$

Hurdle Model

- A hurdle model incorporates 0s using sequential decision making.
- π_i is the probability that $y_i = 0$ acts as the first decision
- g_i represents the count distribution given that $y_i > 0$.

- $P(y_i = j) =$

- $k_i =$

Hurdle Model 2

- Assuming g_i is a Poisson distribution with mean μ_i,
- $E[y_i] =$
- $Var(y_i) =$

Mixture Models Summary

- Heterogeneity model is a continuous mixture.
- Latent class, zero-inflated, and hurdle models are discrete mixtures.
- Distinguish between how zero-inflated vs. hurdle model handle 0s.
- All 4 models handle mean < variance, but the hurdle model only can handle mean > variance cases.

Concept Check #2

- Which statements are true?
- I. If the Poisson model $\ln E[Y] = b_0 + b_1 X$ is used and $b_1 = 0$, then X and Y are uncorrelated.
- II. If the Poisson model $\ln E[Y] = b_0 + b_1 X$ is used and $b_1 < 0$, then $E[Y]$ can be negative.
- III. The negative binomial model and zero-inflated model are appropriate to model a count response variable when concerned about the amount of 0s in the actual distribution.
- IV. Zero-inflated and hurdle models accommodate under-dispersion.
- V. Negative binomial models accommodate over- and under-dispersion.

3. Zero-Inflated Illustration

- A zero-inflated Poisson model with log link models # of accidents using x_1 (age) and x_2 (MV points) as predictors.
- The Poisson count model component has $b_0 = 0.1, b_1 = 0.02, b_2 = 0.2$.
- The zero-inflated model component (binomial with logit link) that predicts the excess zeros has $b_0 = 0.3, b_2 = -0.4$.
- Find the probability of no accidents under the zero-inflated Poisson model for a 35-year old with 2 points.

3. Zero-Inflated Illustration 2

- Zero-Inflated: $P(y_i = j) = \begin{cases} \pi_i + (1-\pi_i)g_i(0), j = 0 \\ (1-\pi_i)g_i(j), j = 1,2, \ldots \end{cases}$

Example 2

- A zero-inflated model is a mixture of a point mass at 0 with 40% probability and a Poisson distribution with mean 1.
- Calculate $P(Y = 0)$ and its mean and variance.

- What if it were a hurdle model with probability of 0 chosen to be 40%? Calculate k_i, and the model mean and variance.

Example 2 Sol

- Zero-Inflated: $P(y_i = j) = \begin{cases} \pi_i + (1-\pi_i)g_i(0), j = 0 \\ (1-\pi_i)g_i(j), j = 1,2, \ldots \end{cases}$
- $E[y_i] = (1-\pi_i)\mu_i$, $Var(y_i) = (1-\pi_i)\mu_i + \pi_i(1-\pi_i)\mu_i^2$
- Hurdle: $P(y_i = j) = \begin{cases} \pi_i, j = 0 \\ k_i g_i(j), j = 1,2, \ldots \end{cases}$
- $k_i = \frac{1-\pi_i}{1-g_i(0)}$
- $E[y_i] = k_i\mu_i$, $Var(y_i) = k_i\mu_i + k_i(1-k_i)\mu_i^2$

Summary Check

- Poisson linear exponential decomposition, canonical link, mean-variance relationship, exposures incorporation
- Formulas for log-likelihood, score equation, deviance, max-scaled R^2, pseudo-R^2, Pearson residual, deviance residual, overdispersion parameter, likelihood ratio test
- Negative binomial, latent class, heterogeneity model concepts
- How to use hurdle, zero-inflated models